Review export of

Task 2 - Case study: Language models (Submission and Peer feedback)

Reviews written by

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Review for Language models - group 7

Based on your reading of the Case study description and this Task 2 submission, which specific areas of the Case study description should the group focus on elaborating in their revision, and why?

Comment criterion

9 comments

OS Olivia Sommer Droob
22-04-2026 12:14:52

I dont really elieve that you perform the sensitivity anaysis required. I believe you should try to vary multiple parameters not just the hardware. Generally some more visual representation would be nice :)

M Max-Peter Schrøder
27-04-2026 21:21:02

I suspect it's slightly too long (over the character limit). But honestly it's really good, otherwise no comments.

AR Andreas Rattenborg Holm-Hansen
25-04-2026 08:25:25
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Unsupported/overstated conclusion. The table only shows compliance under your chosen allocation method and assumptions, not “absolute” sustainability.

YB Yannick Brot Christensen
23-04-2026 12:16:04

I think you should explain your safe operating space part a bit more, especially how you use cost to allocate emissions, because right now it gives some strange results where higher emissions still look equally sustainable. It would help to reflect a bit more on what this actually means and if it affects your conclusion that SLMs are sustainable.

V Viktor Manuel Guijarro
28-04-2026 13:43:53

I think you could work on a sensitivity analysis. You could look through each scenario parameter across a wide range of values rather than just comparing three fixed configurations. But overall really good report.

L Levente Murgas
26-04-2026 14:38:47

I think you got the life cycle stages break down wrong. You needed to divide both the training and prompting into respective substages like data loading and forward pass, backward pass, validation for example for training and measure the emissions with CodeCarbon per these stages.

L Levente Murgas
26-04-2026 14:44:49

There's also a conceptual issue with your scenarios. Check the task I specification for scenarios. You needed to define a baseline, from which additional scenarios are a single parameter varied accross a range of values.

I give you an example, it also took us long before everyone got it in the group.

Let's say your baseline is the same:
4/4/128/GPU

then scenario 1 is when you change only the first parameter:
8/4/128/GPU
16/4/128/GPU
32/4/128/GPU

and each if these experiments gives you emissions and power usage.

then you move onto scenario 2 where you only change the first parameter, while keeping the rest fixed to their baseline and so on.

1 reply

L

Levente Murgas
26-04-2026 14:46:22

This is important so that you can do sensitivity analysis

CJ

Christian Jessen Byskov
28-04-2026 13:50:14

Grandfathering approach

The allocation logic has a bit of a circular problem. So, by using the cost of each run to determine the environmental budget, more expensive scenarios automatically get a larger allowance. This means Alternative A and B don't look worse even though they emit more, which kind of defeats the purpose. You even point this out yourselves! The FU should be the anchor for the allocation instead.

Economic Impacts

Training costs are based on AWS cloud pricing, but inference costs use Danish household electricity rates. Since the whole framing is cloud deployment, it would make more sense to price inference the same way maybe?

Social Impacts

The impacts you describe (noise, air, water, jobs) are solid, but the task specifically asks you to identify stakeholder groups and tie the risks to them. Right now it reads more as a general list of effects rather than "these specific groups are affected in these specific ways." Just adding a brief stakeholder mapping do a lot.

VB

Vlad Burlacu
27-04-2026 09:41:15

I suggest that you revise the Social Impacts and Human Health section, and more specifically, identifying the stakeholder groups. You discuss different aspects of the impact (noise/air/water pollution) but 'locals' and 'workers' are treated as broad categories without substantial differentiation. In addition, you lack a discussion on the privacy or social risk associated with information processed by LLMs/SLMs.

1 reply

AD

Aslan Dalhoff Behbahani
27-04-2026 11:56:46

That is actually a great point, thank you, we will look into it :)

How could the group strengthen the integration and combination of the methods taught in order to better support their sustainability assessment?

Comment criterion

9 comments

OS

Olivia Sommer Droob
22-04-2026 12:17:27

You unfortunately didn't attach your Task 1 so i can't refer to your inference FU. But I assume you have defined a fixed amount of inferences you are comparing, so the comparison of N prompts is not a very strict and comparable way to do LCA. The more course compliant version would be to define a fixed amount of prompts and then vary other paramters (eg max output length)

M

Max-Peter Schrøder
27-04-2026 21:21:35

They already use the methods well, no comments.

AR

Andreas Rattenborg Holm-Hansen
25-04-2026 08:23:42

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Duration and hardware utilization also contribute

YB

Yannick Brot Christensen
23-04-2026 12:15:24

For the social analysis, using a more structured framework like the UNEP S-LCA Guidelines would help make your stakeholder impacts more systematic and clearly connected to the rest of your assessment.

V

Viktor Manuel Guijarro
28-04-2026 13:43:54

Overall, the report is very strong. The only improvement I would suggest is adding more graphs to enhance the visual presentation and make the insights easier to grasp.

L

Levente Murgas
26-04-2026 14:48:17

I think you mixed up grandfathering with FCE, which are 2 different allocation methods.

L

Levente Murgas
26-04-2026 14:48:57

Grandfathering would have assigned each scenario a share proportional to its current share of actual emissions within the sector, not its share of spending.

CJ

Christian Jessen Byskov
28-04-2026 13:51:44

Actually a short paragraph or summary table comparing your scenarios across all three dimensions at the end would give a lot in terms of assesing the complete picture.

VB Vlad Burlacu
27-04-2026 09:52:28

A core imporvement could be done in LCC analysis of the economic dimension. I suggest using a sensitivity plot to illustrate how the costs change (as opposed to two fixed values of N) as the inference volume increases. This could provide a better overview of the exact point where alternative B becomes too expensive compared to your baseline.
Another suggestion would be to improve the vastly descriptive Social Impacts section by adding, for example, metrics for water scarcity (liters per kWh, for instance) because this would allow you to quantify the footprints.